

Exercise 11

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Problem 1a.

Solution. 1. $T\#\mu = 2\text{Leb}_{\mathbb{R}}$. True

2. False

3. $T\#\mu = 4\text{Leb}_{[0,1]}$ True

4. $T\#\mu = \frac{1}{4}\text{Leb}_{\mathbb{R}^2}$ True

5. True

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Problem 1b.

Solution. 1. true

2. true

3. true

4. false

5. true

■

Problem 2. A measure ν on $(E, \mathcal{E})$ is diffuse if

$$\forall x \in E, \nu(\{x\}) = 0$$

and we call it simple if $\nu(\{x\}) \leq 1$.

Assume μ is diffuse, and let $(E_i)_{i \in \mathbb{N}}$ be a partition of E s.t E_i is measurable and satisfies $\mu(E_i) < \infty$. Let M be a Poisson Point Process on $(E, \mathcal{E})$ with intensity μ .

Claim 2a. Show that the restricted measures $\mu_1, \mu_2, \dots$ are diffuse and $M_{E_1}, M_{E_2}, \dots$ are Independent Poisson Processes.

Proof. $\forall i \in \mathbb{N}, \forall x \in E_i, \mu_i(\{x\}) = \mu(\{x\}) = 0$. Thus μ_i is diffuse. Since $E_i \cap E_j = \emptyset$ if $i \neq j$ since $(E_i)_{i \in \mathbb{N}}$ is a partition, thus M_{E_i} are Independent Poisson Point Processes because of Theorem 7.16 (Restriction Theorem). $\square$

Claim 2b. Show that for all $i \in \mathbb{N}$, M_{E_i} is simple almost surely

Proof. By Theorem 3.10, $M_{E_i} = \sum_{i=1}^{Z_i} \delta_{X_i}$ where $Z_i = \text{Poisson}(\mu(E_i))$ and $X_i \sim \mu(\cdot)/\mu(E_i)$.

$$\begin{aligned} \mathbb{P}(M_i \text{ is not simple}) &= \sum_{n \geq 2} \mathbb{P}(Z_i = n) \mathbb{P}(M_i \text{ is not simple} | Z_i = n) \\ &= \sum_{n \geq 2} \mathbb{P}(Z_i = n) \mathbb{P}(\exists a \neq b \ni X_a = X_b | Z_i = n) \end{aligned}$$

For all $a, b \in \mathbb{N} \ni a \neq b$,

$$\begin{aligned} \mathbb{P}(X_a = X_b | Z_i = n) &= \frac{1}{\mu(E_i)^2} (\mu \otimes \mu) \{(x, x) | x \in E_i\} \\ &= \int_{E_i} \mu(\{x\}) \mu(dx) \\ &= 0 \end{aligned} \quad (\text{Since } \mu \text{ is diffuse.})$$

Thus

$$\mathbb{P}(\exists a \neq b \ni X_a = X_b | Z_i = n) = 0$$

Thus $\mathbb{P}(M_i \text{ is not simple}) = 0$. Thus M_i is simple almost surely. $\square$

Claim 2c. Deduce that M is almost surely simple.

Proof. By superpositions,

$$M = \sum_{i \in \mathbb{N}} M_{E_i}$$

Since $E_i \cap E_j = \emptyset$ if $i \neq j$, overlaps only occur in the same E_i . However for each E_i , M_{E_i} is simple almost surely, so no two points in M can coincide. Thus M is simple. The events

$$\{M \text{ isn't simple}\} \subseteq \bigcup_{i \in \mathbb{N}} \{M_{E_i} \text{ isn't simple}\}$$

Thus $\mathbb{P}(M \text{ isn't simple}) \leq \mathbb{P}(\bigcup_{i \in \mathbb{N}} \{M_{E_i} \text{ isn't simple}\}) = \sum_{i \in \mathbb{N}} \mathbb{P}(M_{E_i} \text{ isn't simple}) = 0$.
Thus M is simple almost surely. $\square$

Problem 3a. Let M be a Poisson Point Process on $\mathbb{R} \times [0, \infty)$ with intensity measure $\mu = \text{Leb}$. Is this process almost surely simple?

Proof. $\forall x \in \mathbb{R} \times [0, \infty)$

$$\mu(\{x\}) = 0$$

Thus by 2, M is almost surely simple. $\square$

Problem 3b. Let M be a Poisson Point Process on $\mathbb{R}$ with intensity measure μ given by $\mu(B) = |B \cap \mathbb{Z}|$ for $B \in \mathcal{B}(\mathbb{R})$. Is the process almost surely simple?

Proof. $\forall x \in \mathbb{Z}$,

$$M(\{x\}) \sim \text{Poisson}(\mu(\{x\})) = \text{Poisson}(1)$$

Thus $\mathbb{P}(M(\{x\}) \leq 1) = 2e^{-1}$.

M is simple if (not an iff) $\forall x \in \mathbb{Z}$, $M(\{x\}) \leq 1$. Because of independence of disjoint sets (Property of i of Poisson point process).

$$\begin{aligned} \mathbb{P}(M \text{ is simple}) &\leq \prod_{x \in \mathbb{Z}} \mathbb{P}(M(\{x\}) \leq 1) \\ &= \prod_{x \in \mathbb{Z}} \frac{2}{e} \\ &= 0 \end{aligned}$$

The last equality comes from the fact that $\epsilon > 2 \implies \frac{2}{\epsilon} < 1$. Thus M is not almost surely simple. $\square$

Problem 4. Let M be a ppp($\mu = \lambda \text{Leb}_{\mathbb{R}^d}$), for $\lambda > 0$, on $\mathbb{R}^d$. Let $B_r := \{x \in \mathbb{R}^d : |x| \leq r\}$

Claim 4a. Consider the map $T : \mathbb{R}^d \rightarrow [0, \infty)$ where $x \rightarrow |x|$. Determine the intensity measure of $T\#M$

Proof. $\forall 0 \leq a \leq b$, $(T\#\mu)([a, b]) = \mu(B_b \setminus B_a) = \lambda \text{Leb}_{\mathbb{R}^d}(B_b) - \lambda \text{Leb}_{\mathbb{R}^d}(B_a) = \lambda v_d(b^d - a^d)$ where $v_d := \text{Leb}_{\mathbb{R}^d}(B_1)$ or the volume of unit d-ball. $T\#\mu$ is a σ finite measure. Thus the intensity of $T\#M$ is $T\#\mu$. $\square$

Claim 4b. Prove that a.s

$$\lim_{r \rightarrow \infty} \frac{M(B_r)}{|B_r|} = \lambda$$

where $|B_r|$ is volume of B_r .

Proof. Let $(r_k)_{k \geq 0}$ where $|B_{r_k}| = k$. For $k \geq 1$:

$$X_k := M(B_{r_k} \setminus B_{r_{k-1}}) \sim \text{Poisson}(\mu(B_{r_k} \setminus B_{r_{k-1}})) = \text{Poisson}(\lambda)$$

$$S_n := \sum_{i=1}^n X_k = \sum_{i=1}^n M(B_{r_i} \setminus B_{r_{i-1}}) = M(B_{r_n}) \sim \text{Poisson}(\lambda n)$$

By the Strong Law of Large Numbers:

$$\Pr \left(\lim_{n \rightarrow \infty} \frac{S_n}{n} = \lambda \right) = 1$$

Substitute $S_n := M(B_{r_n})$ and $n = |B_{r_n}|$.

$$\lim_{n \rightarrow \infty} \frac{M(B_{r_n})}{|B_{r_n}|} = 1$$

a.s. $\forall q > 0, \exists n \in \mathbb{N} \ni r_n \leq q \leq r_{n+1}$. By Monotonicity of M and $|\cdot|$, $M(B_{r_n}) \leq M(B_q) \leq M(B_{r_{n+1}})$ and $n = |B_{r_n}| \leq |B_q| \leq |B_{r_{n+1}}| = n + 1$. Thus

$$\frac{M(B_{r_n})}{|B_{r_{n+1}}|} \leq \frac{M(B_q)}{|B_q|} \leq \frac{M(B_{r_{n+1}})}{|B_{r_n}|}$$

Note $\frac{M(B_{r_n})}{|B_{r_{n+1}}|} = \frac{M(B_{r_n})}{n+1} = \frac{n}{n+1} \frac{S_n}{n}$ and $\frac{M(B_{r_{n+1}})}{|B_{r_n}|} = \frac{M(B_{r_{n+1}})}{n} = \frac{S_{n+1}}{n+1} \frac{n+1}{n}$.

Thus

$$\frac{n}{n+1} \frac{S_n}{n} \leq \frac{M(B_q)}{|B_q|} \leq \frac{n+1}{n} \frac{S_{n+1}}{n+1}$$

Almost surely,

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} \frac{S_n}{n} = \lambda = \lim_{n \rightarrow \infty} \frac{n+1}{n} \frac{S_{n+1}}{n+1}$$

By the squeeze Theorem, a.s

$$\lim_{r \rightarrow \infty} \frac{M(B_r)}{|B_r|} = \lambda$$

□